KNIME demo: Speaker script

Tax team session · 20 slides · about 30–35 minutes

Fill in the [__] timings on slide 15 and replace the bracketed Alteryx note on slide 13 before presenting. *Italic text in brackets* is a stage direction, not for reading aloud.

1 Cover

Good morning everyone, and thanks for joining. Over the last few weeks I've been running a proof of concept on KNIME, a data preparation and automation tool, as part of our evaluation against Alteryx. Today I'll keep it practical. I'll show you what KNIME can do and where it falls short, what it costs, the workflow I built running live, and how you can install it on your own laptop. Please stop me any time with questions.

2 Agenda

We'll go through four parts. First, features, capabilities and limitations. Second, open source and licensing, so what's free and what's paid. Third, a live demo of the POC. And finally, how to set it up on your machine. Each part is about eight to ten minutes, and we'll keep time for questions at the end.

3 Section 01 – Features, capabilities and limitations

Let's start with what KNIME actually is and what it can do.

4 KNIME turns repetitive data work into a visual, reusable workflow

The simplest way to explain KNIME is this. Think of any Excel process you repeat every month: copy-paste from a few files, VLOOKUPs, filters, a pivot at the end. In KNIME, each of those steps becomes a box called a node. You connect the nodes left to right, and that picture becomes your process. Once it's built, next month you just point it at the new file and press Execute. Most of this needs no coding. It runs on your desktop, so the data stays on your laptop.

5 Core capabilities

Here are six things KNIME does well.

- Connect: it reads almost anything, including Excel, CSV, databases and cloud storage.
- Clean: removing duplicates, fixing date formats, handling blanks, which is what many of us do by hand today.
- Join and reconcile: matching two sources and showing what doesn't tie. For us, that's probably the most relevant one.
- Summarise: group, pivot and write the result back to Excel.
- Code when needed: if you know Python or SQL you can drop it in, but you don't have to.
- K-AI, a built-in AI assistant. One caution here: it sends prompts to an external AI service, so we'd need to check policy before using it with client data.

6 Where it could help the Tax team

So what does this mean for Tax? These are candidate use cases. I haven't built them yet, and I'd really like your input on which one hurts most today.

- GST or TDS reconciliation: matching our books against GSTR-2B or 26AS and listing the mismatches.
- Consolidating Excel files from multiple entities into one table.
- Cleaning GSTINs, PANs and vendor names before analysis.
- Tying out ledgers to returns.
- Month-end repeatability: the same steps every month on new files.

The last point is the audit trail. The workflow itself shows exactly what was done to the data, which is much clearer than a heavily edited spreadsheet.

7 What you'll notice when you use it

To be balanced, here's what you'll notice when you actually use it.

- Learning curve: there are thousands of nodes, often several that do nearly the same thing, so beginners spend time finding the right one.
- Formula syntax: different nodes use different formula styles. In my POC, a simple "if the date is in this format, convert it" needed three nodes, where Alteryx does it in one formula.
- Memory: it's a Java application, and on large files you need to raise its memory setting.
- Scheduling: the free version only runs when you press the button. Automatic scheduled runs need the paid Hub.
- Interface: it's a little less polished than Alteryx.
- Governance: on desktop, workflows sit on individual laptops, so version control is our responsibility.

8 Section 02 – Open source and licensing options

Now the question everyone asks: what does it cost?

9 Desktop KNIME is free, not a trial version

The good news is that the desktop tool is completely free. It's not a trial or a cut-down version. There's no licence fee, no user limit, no expiry, and no limit on data size. It's open source under the GPLv3 licence, which allows commercial use. KNIME makes its money from an optional server product called KNIME Hub, not from the desktop tool. One caveat: even though it's free, we still need IT approval to install it on firm laptops.

10 Licensing options

Here's the full picture. The top row, the desktop Analytics Platform, is what all of us would use day to day, and it's free. Everything below it is about sharing and automating.

- Pro and Team are cloud plans: 19 euros a month for one person, or 99 euros a month for a team of three.
- Business Hub comes in Basic, Standard and Enterprise editions, priced on request based on computing capacity and number of users. The Enterprise edition can be installed on our own infrastructure, which matters for client data.

These prices are from KNIME's website as of this month, and they can change.

11 Which option fits us

My suggestion is simple: we don't need to spend anything to start. Everyone installs the free desktop version, and we pick one or two real Tax processes to try. Only once we see real value, and need things like scheduled monthly runs or central access control, do we look at Business Hub. That way any spending decision is based on actual usage, not assumptions.

12 Section 03 – Demo of the POC

Let me now show you the proof of concept.

13 A 20-node data-prep and reconciliation workflow

I built a workflow with 20 nodes. It reads three files (orders, products and customers), cleans the data, keeps only completed orders, joins the three files together, calculates revenue, and writes three summary sheets to Excel. I used sales data instead of tax data so there are no confidentiality concerns, but the pattern is exactly a reconciliation: multiple sources, clean, match, calculate, summarise. [If you built it in Alteryx too, say so here.]

14 Real-world data problems the workflow handles

I made the data deliberately messy, because real data always is. There were 46 duplicate rows, and about one in ten dates were in a different format. 194 rows had no quantity, region names were typed inconsistently, like NORTH in capitals versus North, and some orders were cancelled or returned. Each problem is handled by a specific node, shown on the right. These are exactly the problems we see in tax data: GSTIN typos, mixed invoice-date formats, blank fields and duplicates.

15 Results tie out to an independent calculation

Before trusting any tool, I worked out the expected answers separately. The KNIME output matched: 7,819 clean, matched rows out of about 10,000 raw rows, and the same revenue total. I also ran it on a one-million-row file, which produced about 784,000 matched rows. On speed, KNIME took [__] seconds for 10,000 rows and [__] seconds for a million rows, and building the workflow took about [__]. For a Tax team, "does it tie out" is the question that matters, and here it does.

16 Live demo

(Switch to KNIME.)
Let me show you this live. Here's the workflow. When I click Execute All, watch the nodes turn green one by one. Red means not configured, amber means ready, and green means done.

(Right-click the Duplicate Row Filter.) Here's the output after removing duplicates: exactly 10,000 rows.
(Open the second Joiner.) After the joins we have 7,819 rows.
(Open the totals node.) And here's the reconciliation total.

Now the part that shows the real value. *(Open the first CSV Reader and switch to the 1M file.)* I just change the input file and re-run. Same logic, a million rows, no rework.

(Open the Excel output.) And here's the final output in Excel.

17 Section 04 – How to set up KNIME on your machine

Finally, how you can get this running yourselves.

18 Six steps to get KNIME running

It's six steps.

1. Raise an IT software request. It's free, but it still needs approval.
2. Download it from knime.com/downloads. If you don't have admin rights, use the zip version, which you just extract and run.
3. Install it and choose a workspace folder where your workflows will live.
4. This is the one people skip: open the knime.ini file in the install folder and raise the memory setting to about half your RAM. On a 16 GB laptop, that's 8 GB. Otherwise large files will be slow.
5. Add any extensions you need from the menu. On the office network this may be blocked, in which case IT can give us an offline package.
6. I'll share the POC workflow file. Import it and press Execute All.

19 What your machine needs

The requirements are light. It runs on Windows 10 or 11, and on Mac and Linux too. 16 GB of RAM is comfortable for large files, and 8 GB is fine for smaller work. You don't need admin rights with the zip version. The latest version is 5.12. For a team rollout, I'd suggest we all use the long-term support version so everyone is on the same release.

20 Next steps

So, next steps. This week, install KNIME and run the POC workflow I'll share. Over the next two weeks, let's pick one real Tax process to pilot, and GSTR-2B matching would be a good candidate. Then we compare the time saved and use that to decide between KNIME and Alteryx. Which process would you pick to try first? And I'm happy to sit with anyone who gets stuck installing. Questions?